

## PATTERN MATCHING

### Aim:

To write a Prolog program to implement **pattern matching** between two lists.

### Objective:

To check whether a given pattern matches a sequence of elements using Prolog's **unification and recursion**.

### Algorithm:

1. ☐ Start the program.
2. ☐ Read the pattern and input list.
3. ☐ Compare the first element of the pattern with the first element of the input.
4. ☐ If they match, recursively compare the remaining elements.
5. ☐ If both lists become empty, the pattern matches successfully.
6. ☐ If any element does not match, return false.
7. ☐ Stop.

## Prolog Program:

% Pattern Matching Program

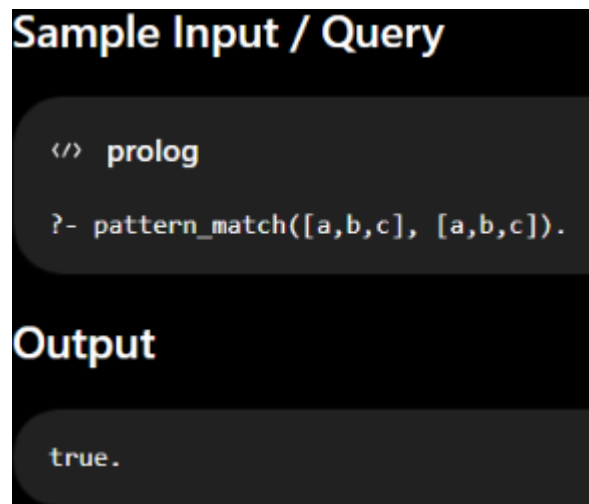
```
pattern_match([], []).
```

```
pattern_match([H1|T1], [H2|T2]) :-
```

```
    H1 = H2,
```

```
    pattern_match(T1, T2)..
```

## Output

A screenshot of a Prolog interpreter interface. It has a dark background with light-colored text. At the top, it says "Sample Input / Query". Below that, in a light gray box, is the Prolog code: "</> prolog" followed by a new line and "?- pattern\_match([a,b,c], [a,b,c]).". Below this, it says "Output". At the bottom, in a light gray box, is the output: "true.".

```
Sample Input / Query

</> prolog

?- pattern_match([a,b,c], [a,b,c]).

Output

true.
```

## Result:

Thus, the Prolog program successfully implements **pattern matching using unification and recursion**.